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REVIEW
Thromboembolic Diseases

Rationale for and approach to establishing a multidisciplinary acute pulmonary embolism expert care team

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Patients with acute pulmonary embolism (PE) may present with cardiac arrest, overt or impending cardiogenic shock and/or severe respiratory insufficiency. Immediate evaluation and management of these patients require high clinical suspicion along with (bedside) imaging to confirm the diagnosis, targeted haemodynamic and/or respiratory support, appropriate anticoagulant treatment, and in many cases reperfusion therapy. The immediate treatment decision-making is largely driven by local expertise and resources and should be guided by the individual patient's characteristics such as cardiopulmonary comorbidities, risk of bleeding and location, extent and haemodynamic impact of the clot. Over the past years,

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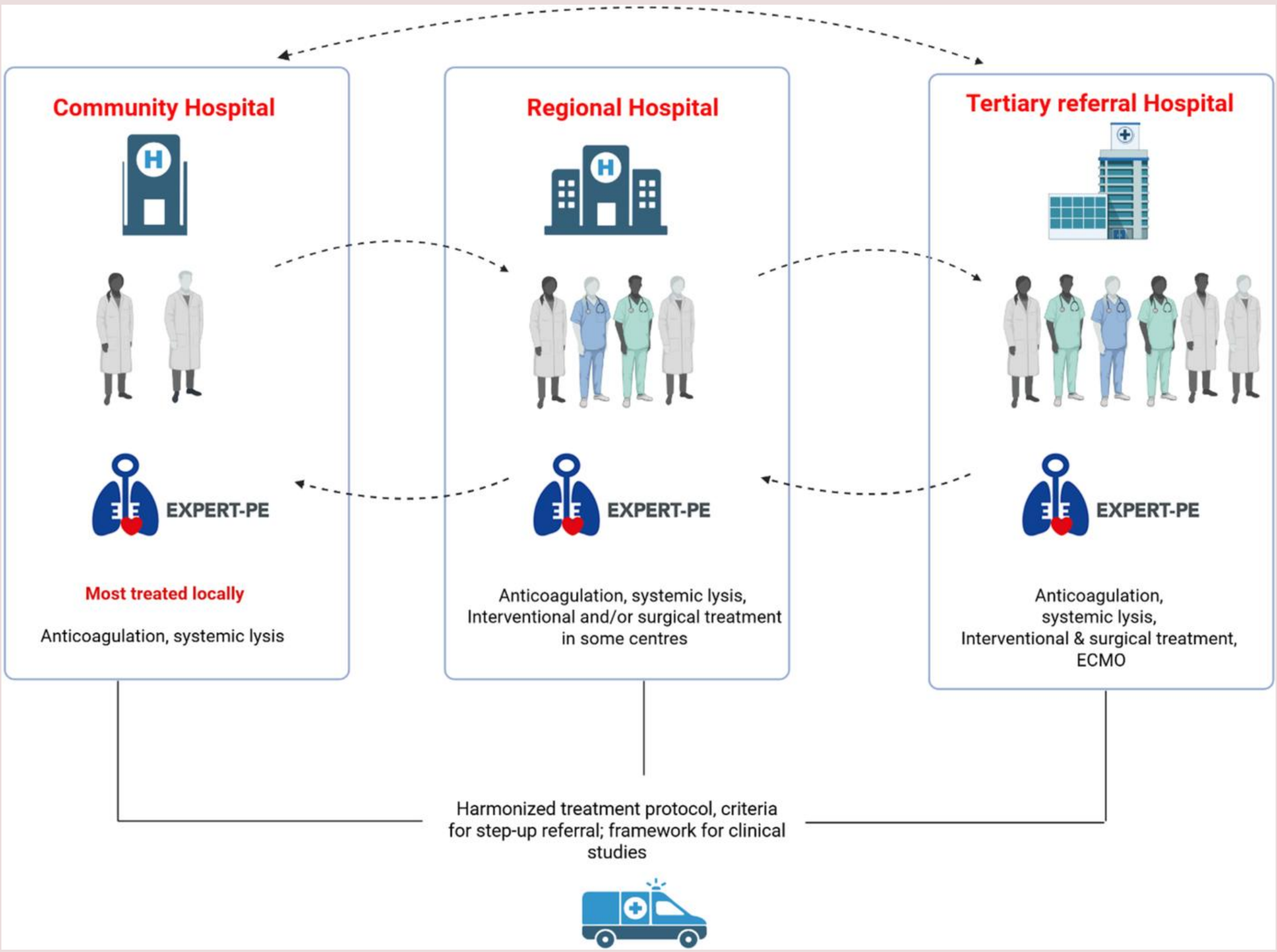
treatment options for patients with severe PE have expanded substantially. For instance, several new catheter-guided reperfusion therapies have emerged and experience with circulatory mechanical support has increased. Along with the rise of new interventional therapies has come the introduction of expert multidisciplinary pulmonary embolism (EXPERT-PE) care teams, composed of multidisciplinary specialists involved in treating severe acute PE. This model of care provides a platform for rapid decisions on individualized treatment strategies, combining expert opinion from all involved specialties, setting the quality standards for modern local and regional equity PE care, and forming the base for future research in this area. Clinical decisions should be evidence-based where possible, and incorporate the individual patient's and their carer's preferences, values, and priorities, as well as those of the managing clinicians and care team. In this review, we summarize the evidence for the introduction of EXPERT-PE care teams and provide a practical manual for their successful implementation.

Graphical abstract



Keywords Pulmonary embolism • Interdisciplinary • Europe • Therapy • Diagnosis

Introduction

In the past decade, multidisciplinary team-based care has rapidly gained recognition as a promising care model for patients with acute pulmonary embolism (PE), particularly for those with a more severe disease presentation. Formally defined in its structure in the United States, this care model is now integrated more and more in hospitals around the globe.[1–4] The general concept is that by bringing together *ad hoc* the experts from all disciplines involved in PE diagnosis and management in a given institution, EXPERT-PE care teams provide a real-time platform for prompt sharing of relevant clinical information with the aim of facilitating, accelerating and optimizing decision-making in the flexible timeframe appropriate for this acute setting. The need for this care model emerged from epidemiological data underlining persistently high mortality among patients with severe PE,[5] as well as technical advancements in PE care, notably catheter-directed reperfusion (CDR) that has drawn increasing interest from multiple medical specialities, but for which the evidence on selection and application is yet uncertain. However, the involvement of different disciplines, each with its own perspectives and protocols, has added complexity to medical decision-making in daily practice.[6–10]

Organizing dedicated multidisciplinary care for patients with PE around the clock, however, requires considerable effort and allocation of resources, both in personnel and in information and communications technology support, as well as local or regional collaboration beyond the boundaries of medical specialities. Moreover, it often necessitates bridging of competing medical and/or scientific interests.

Addressing the optimal management of severe presentations of acute PE is a critical mandate in contemporary healthcare, necessitating a collaborative focus on patient-centred care and evidence-based practices. This includes providing comprehensive guidance on haemodynamic, respiratory, and laboratory monitoring, individual risk assessment using validated scoring systems, haemodynamic support, advanced reperfusion modalities and their appropriate selection and timing, and the judicious use of anticoagulation. Adequate training for healthcare professionals, particularly in bedside tools such as echocardiography, and developing standardized clinical pathways and protocols are essential components of this effort. This clinical review, prepared by members of the EXPERT-PE study group of the Acute Cardiovascular Care Association (ACVC) of the European Society of Cardiology (ESC) and other European scientific societies relevant to this topic, outlines the current evidence-based approach to multidisciplinary care for patients with acute PE and cardiopulmonary compromise. This document seeks to establish an operational framework to support the introduction and implementation of multidisciplinary EXPERT-PE care teams in hospitals across Europe. By fostering the adoption of EXPERT-PE care teams, we aspire to facilitate creating and coordinating a robust network of EXPERT-PE care teams across Europe. This network will play a pivotal role in harmonizing PE management, providing opportunities for clinical education, and fostering collaboration on research agendas. By prioritizing follow-up protocols, identifying research priorities, and promoting evidence-based practices, the study group aims to drive innovation in PE care, ultimately improving the quality and outcomes of care while setting a global benchmark for managing this life-threatening condition.

Rationale of multidisciplinary care for acute PE

Management of acute, life-threatening conditions in critical care usually requires a coordinated, multidisciplinary approach that draws on all available expertise related to the specific condition, including relevant treatment options and management of potential complications. In the setting of severe acute PE, this includes foremost experts in the management of haemodynamic and severe respiratory insufficiency, bedside and radiographic imaging, anticoagulation and (pharmacological, interventional or surgical) reperfusion therapy, depending on local resources and experience.[11–13] The concept of multidisciplinary care for critically ill patients has been widely established, for instance by cardiogenic shock teams.[14–17] Several single-centre studies initially established the proof of concept, suggesting lower in-hospital mortality after implementation of a cardiogenic shock care team.[15,16] For example, an analysis of over 1200 patients with cardiogenic shock treated within the Critical Care Cardiology Trials Network showed that mortality in hospitals with an institutional multidisciplinary care team was 30% lower than in hospitals without such dedicated care teams. Also, hospitals with cardiogenic shock teams utilised invasive monitoring more often and had lower rates of use of mechanical cardiac support, though when chosen, tended towards more advanced forms, such as micro-axial flow pumps, rather than intra-aortic balloon pumps.[17] Put together, this might suggest that decision-making is altered by the presence of a dedicated multidisciplinary team.

Acute PE may present as a life-threatening cardiovascular condition, but the majority of patients can be successfully managed with anticoagulation alone and even managed at home in up to 40% of cases diagnosed at emergency departments.[18,19] At the other end of the PE severity spectrum are patients at high risk of haemodynamic or respiratory decompensation with signs of acute right ventricular (RV) pressure overload and dysfunction, unstable vital parameters and/or signs of organ failure, or those in shock with overt multisystem organ failure or even cardiac arrest.[12,13] Mortality in these patients can reach 70% depending on the stage of shock—and therefore require immediate haemodynamic and respiratory support, appropriate anticoagulant treatment and often reperfusion.[8–13,20–22] Reperfusion therapy for acute PE may consist of systemic thrombolysis, surgical embolectomy or CDR, each requiring specific expertise and a holistic assessment of the patient to determine the optimal approach.[8–10,12,23–25] Decision-making regarding reperfusion is guided by the haemodynamic status of the patient; for instance, cardiac arrest or refractory shock may require extracorporeal circulatory assistance using veno-arterial extracorporeal membrane oxygenation (ECMO) before reperfusion treatment is initiated, especially in the setting of surgical or percutaneous treatment.[8,12,21,25–29] Other details of presentation matter too: for patients with clot-in-transit trapped in a patent foramen ovale urgent open surgery may be the best option; alternately, patients with a high risk of bleeding or ongoing bleeding are not good candidates for systemic thrombolysis.[9,10,12,30] Also, patients with acute-on-chronic PE and pre-existing severe pulmonary hypertension present different challenges that must be recognized at the time of diagnosis. If teams are considering CDR, choice of technology and technique must take account of the altered pathophysiological and risk profile, location of the clots and risk of bleeding.[31,32] Hence, the planning of reperfusion treatment is critically dependent on the availability of imaging to confirm the diagnosis, accurately assess the presence of deep vein thrombosis (DVT) or clot-in-transit, and the differentiation between acute and acute on chronic PE.[9,12,33–35] The choice of anticoagulation, still the cornerstone of treatment of all patients with PE, may also be challenging as the severity of shock, patient characteristics such as bleeding risk, body-mass index and renal function as well as the chosen reperfusion approach determine the optimal anticoagulation regimen and appropriate monitoring plan.[9,36–38] Lastly, as PE occurs frequently in elderly patients or those with life-limiting comorbidities such as advanced cancer or severe renal insufficiency, *ad-hoc* geriatric assessment and determination of the limits to appropriate care for individual patients may be needed as part of the decision making process.[39–41] After the initial treatment has been applied, patients require ongoing evaluation of the treatment response, and if the initiated treatment fails to reverse signs of haemodynamic compromise or respiratory insufficiency, escalation of care may be decided on.[9,11,12,42] All in all, the acuity and complexity of these so-called 'high-risk' or 'massive' PE patients clearly requires an interdisciplinary, collaborative, and standardized team-based approach to management. For that reason, the European Society of Cardiology guidelines on management of acute PE—endorsed by the European Respiratory Society—encourage the set-up of multidisciplinary teams for management of high-risk and selected cases of intermediate-risk PE.[12] The concept of multidisciplinary teams for PE management is currently spreading across Europe.[43,44]

Evidence for EXPERT-PE

A growing body of observational literature suggests that implementation of a dedicated institutional multidisciplinary care pathway for (severe) acute PE may be associated with better outcomes of care.[45–50]

The first report dates back to 2016, in which the initial experience with a multidisciplinary PE care team, in the US literature referred to as 'PERT', was reported from the Massachusetts General Hospital in Boston.[45] Over a period of 30 months, PERT was activated 394 times for patients of whom 80% were ultimately diagnosed with acute PE. Patients were classified as submassive PE in 46% and massive PE in 26%. The team proposed anticoagulation alone in 69%, with either catheter or systemic thrombolysis used in the remaining 11% of patients. The overall 30-day mortality was 12%. This single-centre study did not compare the outcomes of care before and after introduction of their PERT, but a later study did, comparing outcomes of 884 patients diagnosed with PE before the establishment of a local PERT, to those 1158 patients with PE post-PERT implementation. Of those 1158 patients, 165 (14%) were selected for a PERT discussion.[50] Overall utilization of reperfusion therapy was similar between groups (5.4% vs. 5.4%, respectively), with decreased use of systemic thrombolysis and increased application of CDR post-PERT implementation. Both PE-related (2.6% vs. 2.7%, respectively) and overall in-hospital mortality were comparable (6.8% vs. 6.2%, respectively), as was the rate of major bleeding (5.8% vs. 7.0%, respectively) though it is important to note that 86% of the post-PERT patients with PE did not receive a discussion and therefore outcomes from the overall cohort could only have been modified by altered strategies in a small minority, albeit of what were likely to be the sickest patients.

The impact of the establishment of a PERT in the Gill Heart and Vascular Institute in Kentucky US, was evaluated by comparing the use of resources and outcomes of care among matched cohorts of patients with submassive or massive PE treated before and after the institution of a local PERT.[46] Matching, based on ICD-10 coding data, was performed with the aim of identifying patients with comparable risk and comorbidity profiles. A total of 992 patients before and 77 after PERT establishment were compared. The mortality in both groups was comparable (13% vs. 15%, respectively). The length of stay was lower in the PERT-treated patients (3 days less, mostly at ICU level) but costs were higher (~\$4600, 37% increase) driven partly by substantially increased use of ECMO and catheter directed treatment (1.7% vs. 7.8%, and 1.8% vs. 40%, respectively).

The PERT programme at the Cleveland Clinic (US) was launched in July 2014. In a retrospective study, the use of resources and outcomes of care of all patients with confirmed PE were compared between the pre- and post-PERT era; patients from the first 6 months after launch of the programme were not taken into account to rule out the impact of a learning curve.[47] A total of 769 patients were included in the study, of whom 426 in the post-PERT era (55%). These latter patients were found to be diagnosed with slightly more severe PE than those in the pre-PERT era, based on the presence of RV dysfunction, positive biomarkers and simplified pulmonary embolism severity index (sPESI) at baseline. The PERT was activated for 13% of all cases. In the post-PERT era, the time-to-therapeutic anticoagulation was ~3.5 h shorter, the use of inferior vena cava filters decreased from 22% to 16%, and the use of advanced strategies including CDR increased from 2.8% to 5.6%. Notably, in the post-PERT era, the rate of major and clinically relevant bleeding decreased by 50% and overall 30-day mortality by 45%, especially in those with intermediate to high-risk PE.

In the Abbott Northwestern Hospital, Minneapolis US, a 2-step approach for introduction of a PERT was followed.[48] First, a hospital-wide treatment algorithm was introduced by a multidisciplinary team that was based on the existing guidelines, current publications, and local expertise and resources. Two years later, a PERT team was implemented that could be activated in complex patients or those at high risk of adverse outcomes or death. In a retrospective study, treatment patterns and outcomes of care were evaluated and compared between the period before introduction of the hospital protocol, where patient management was determined by the attending physician with subspecialty consultation as deemed appropriate, the period after introduction of the protocol but before PERT installation, and thereafter. A total of 1105 PE patients were admitted over the complete period; PERT was activated in 21% of the cases in the last study period. Over time, the proportion of patients treated with reperfusion increased from 4.7%, to 8.2% and 16%, respectively, with a notable decrease of the use of CDR from 60% to 16% and 18%, respectively. Where reperfusion therapy was selected, the choice of systemic lysis (mostly at reduced dose) over other reperfusion methods, increased over time from 40% to 84% and 82%, respectively. Mortality and readmission rates remained unchanged across the 3 study periods, as did length of stay at the ICU, though the total length of stay was reduced from 4.78 to 2.81 days to. The authors concluded that the introduction of the hospital-wide protocol seemed to be the main driver of the changes in treatment use of health care resources over time.

The University of Virginia Medical Center PERT was formally inaugurated in 2017. A retrospective study to compare use of healthcare resources, outcomes and costs before and after the inauguration was undertaken.[49] The historical comparator cohort was composed of 237 patients and the post-PERT cohort consisted of 317 patients, in whom PERT was activated in 120 (38%). After PERT inauguration, the use of CDR increased from 2.9% to 8.2%, while length of stay and overall costs were comparable between the 2 periods. A relative 30% lower mortality was observed after PERT instigation, and 30-day readmission rates were lower.

A meta-analysis that pooled the data of all published observational studies reported that mortality outcomes pre- and post-PERT implementation did not significantly differ (risk ratio 0.89, 95%CI 0.67–1.19), though when analysis was restricted to those patients with intermediate or high-risk PE, the risk ratio was 0.71, albeit with statistically non-significant 95% confidence intervals of 0.45–1.12.[43] In the same study, the use of advanced therapies increased (risk ratio 2.67, 95%CI 1.3–5.5) and in-hospital stay decreased by 1.6 days. Notably, all pooled studies were retrospective, resulting in considerable risk of bias, and both the composition as well as mode of operation and local protocol differed between the studies in the meta-analysis. Understanding how different EXPERT-PE care team programme structures impact clinical care, and if a certain set-up should be preferred over another remains to be investigated.

Organizational framework

The aims for establishing EXPERT-PE care teams can be summarized as follows. First, it can improve awareness and structures for the diagnosis of PE, thereby facilitating effective and early PE diagnoses. Second, appropriate risk-stratification and treatment allocation of patients can be improved by a standardized, multi-disciplinary approach that only EXPERT-PE care teams can provide. Third, it can improve the local structure for performing studies and trials in the field of PE and increase the scientific excellence. Fourth, it provides a safeguard for avoiding inequities in access to cardiovascular care.

To establish EXPERT-PE care teams, we propose a stepwise approach (*Figure 1*). *Step one* should be to seek support from the hospital administration, including a discussion on relevant financial and technical support, followed by an inventory of all local disciplines already involved in the diagnosis and treatment of PE. This includes an inventory of regional hospitals and their local structure and resources with the ultimate aim of building a regional PE network of treating and referring centres, given that not all centres will have availability of all potential treatments described in this document. Another important partner is the regional emergency patient transport service. Relevant local disciplines may include—but are not limited to—acute medicine, intensive care medicine, (interventional) cardiology, (interventional) radiology, haematology, vascular medicine/surgery, angiology, cardiothoracic surgery and pulmonology. Notably, the final structure of EXPERT-PE care

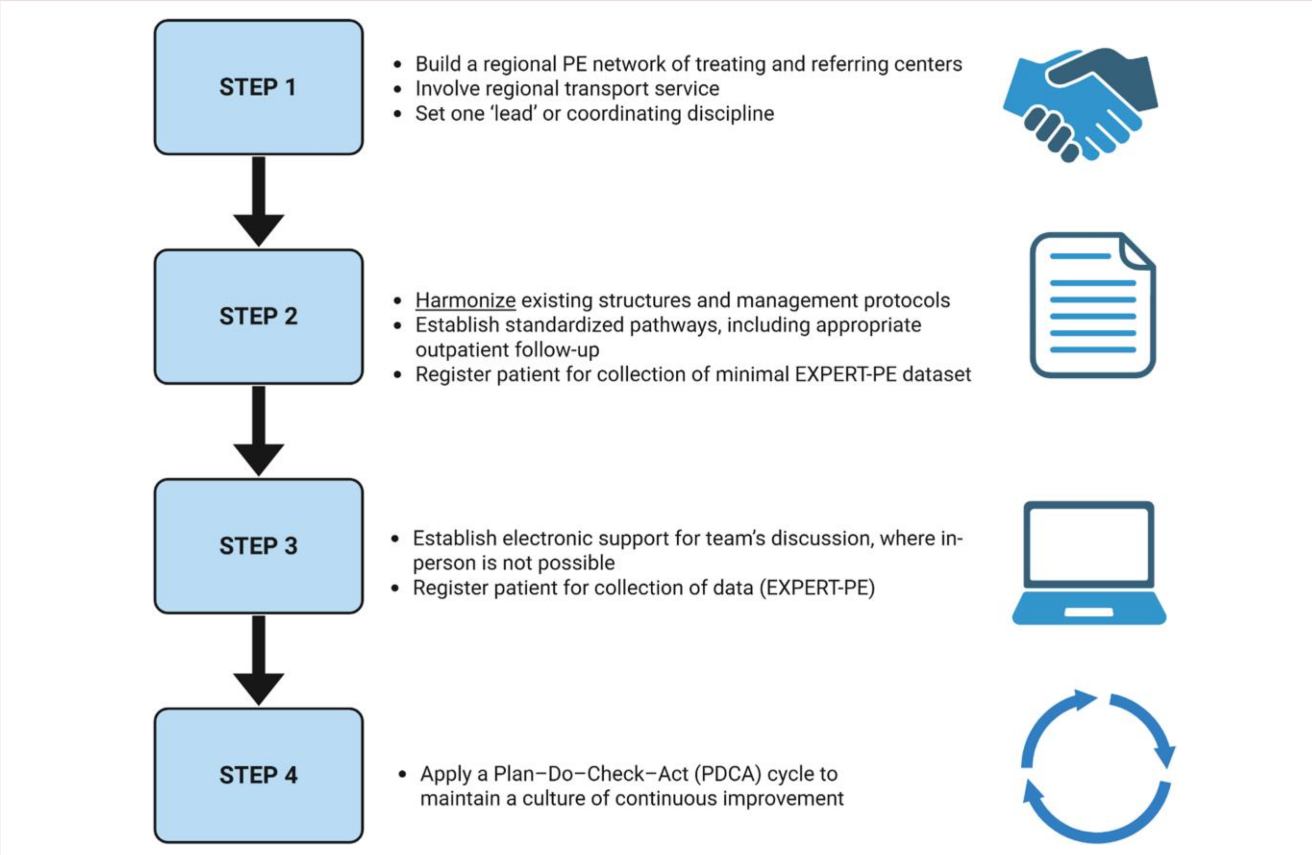


Figure 1 Proposed stepwise approach to establish a local and regional EXPERT-PE care team. Created in BioRender. Ni ainle, F. (2025) https://BioRender.com/ae4xv58

teams varies markedly and should depend on local resources and preferences. One of the keys to successful implementation is to appoint one 'lead' discipline. They have the task to receive all EXPERT-PE calls, initiate and coordinate case discussions, and maintain communication and contact between all involved local and regional partners. Notably, an alternating 'lead' system rather than a single 'lead' discipline might also be suitable in daily practice as it underlines the multidisciplinary character of the team, and allows for a fairly shared workload.

Step two involves reviewing and harmonizing existing structures and management protocols at both local and regional levels (*Figure 2*). Ultimately, a broad consensus should be reached among all involved disciplines and partners. Beyond providing standardized pathways for diagnosing and treating acute PE, ideally for any clinical presentation, these protocols should clearly define:

(1) The specific circumstances and patient profiles for which the EXPERT-PE care team should be activated.
(2) The procedure for activating the care team (see also Step 3).
(3) The core members of the EXPERT-PE care team (should aim to be available for all discussions) and members than can be called upon ad-hoc if deemed necessary, including how 24/7 coverage of relevant disciplines is organized and guaranteed.
(4) The minimum clinical information required to facilitate effective team discussions as well as the preferred available clinical and imaging information, including how these are made available among team members.
(5) The decision-making process, including the initial management strategy (as well as contingency plans B and C), the allocation of responsibilities among specialties for various aspects of the treatment plan, and the oversight of day-to-day patient care during admission.
(6) The standard operating procedures for patient transfer from referring hospitals.
(7) Who is responsible for maintaining and regularly updating the local and regional protocols.
(8) Who is responsible for collecting and presenting regular PERT audit data to the team, the hospital and the network, detailing change and identifying areas for potential improvement (e.g. hospital transfer times).

Importantly, beyond the immediate diagnosis and treatment of individual patients, the EXPERT-PE care team may also take collective responsibility for protocols related to transition to ambulatory care, ensuring appropriate outpatient follow-up, and advancing the team's scientific initiatives and goals.

Step three is to establish reliable electronic support to support the team's discussion, including telemedical/electronic referral and consultation options for network hospitals. A bedside assessment of the patient by the attending members of the EXPERT-PE care team is the preferred basis of the multidisciplinary discussion, but this will not always be feasible. Several mobile technologies are available for real-time online case evaluation, both generic as well as PE-specific. Commercial AI-driven data platforms provide means to automatically integrate clinical, radiological and biochemical data, detect patients with PE and activate the EXPERT-PE care team. Use of such technologies facilitates information sharing and quick decision making in the wider network of hospitals affiliated to the team.[11,44,51,52] Funding for implementing and using these platforms should be part of the initial negotiation with the hospital's administration described in *step 1*.

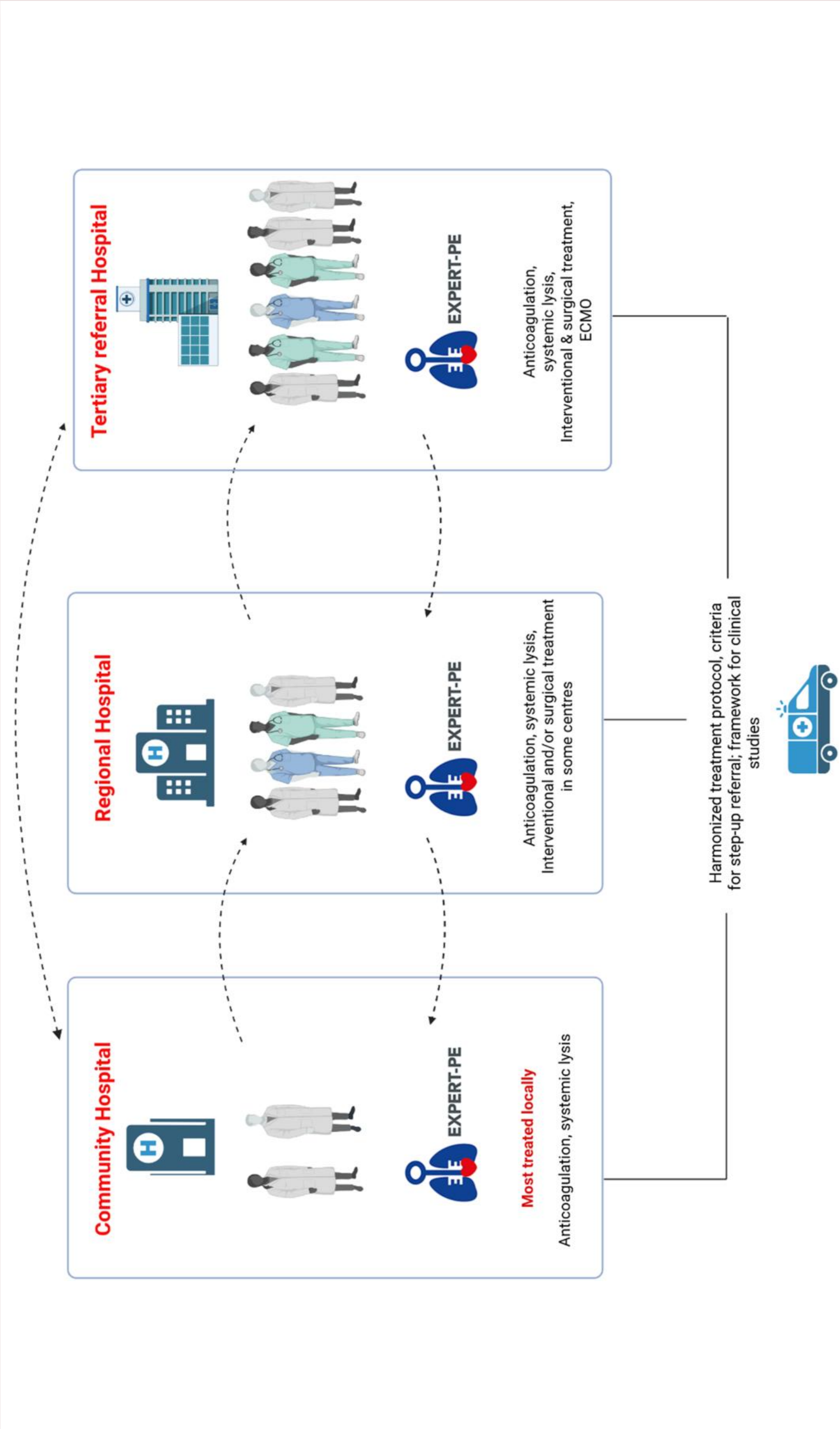


Figure 2 The EXPERT-PE model for regional collaboration in PE care. Created in BioRender. Ni ainle, F. (2025) https://BioRender.com/hrmhgsx

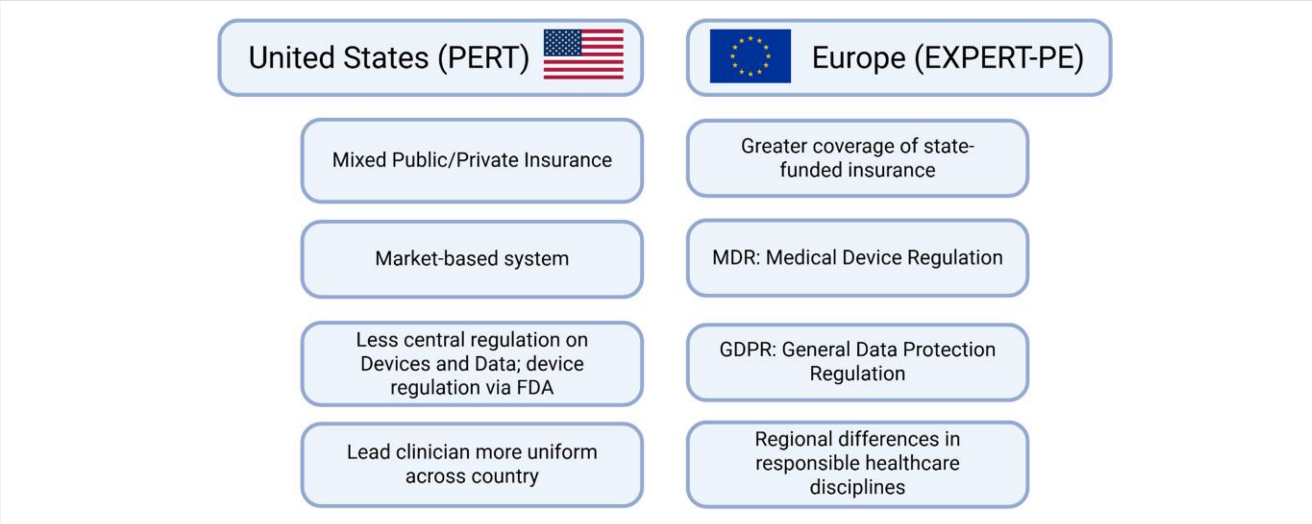


Figure 3 The European perspective of establishing EXPERT-PE care teams. Created in BioRender. Ni ainle, F. (2025) https://BioRender.com/ss0voy5

Step 4 is to apply a Plan–Do–Check–Act (PDCA) cycle to maintain a culture of continuous improvement. Both the logistical, administrative and clinical aspects and outcomes of the consultations should be part of this cycle. It is recommended to involve all local and regional partners in this process.

Quality standards

As timing can be crucial in higher-risk PE, activation of the EXPERT-PE care team should not lead to relevant delays in patient treatment. While there are few scientific data defining optimal timeframes for treatment decisions in acute PE, we propose that activation is to occur as early as possible after a patient meets the predefined criteria for EXPERT-PE activation, and the multidisciplinary discussion should start as soon as possible -ideally within 30 min- making sure that crucial treatment such as initial anticoagulation is immediately applied, pending the team's full treatment proposal. Decision making itself should also be fast and effective following EXPERT-PE activation, with proper and verifiable documentation of decisions. The EXPERT-PE recommendation may involve additional diagnostic steps necessary to ultimately determine the management strategy, but if reperfusion treatment is advised, this should be applied as early as reasonably possible—ideally within 60 min after the decision was taken.[9] These proposed benchmarks align with the treatment timeframes established in other acute conditions such as myocardial infarction and stroke, where 'time-to-treatment' is a recognized determinant of outcome. To ensure quality improvement and enable research, the timing of activation, team discussion, treatment initiation, and any causes of delay should be routinely documented and reviewed. Regular audit of these metrics is a core responsibility of the EXPERT-PE team.

European perspective

The PERT Consortium® is a US based organization dedicated to establishing, optimizing, and expanding multidisciplinary Pulmonary Embolism Response Teams (PERT), and their quality care for PE patients. The PERT consortium initiative was launched in 2015, and involves yearly conferences, frequent newsletters, educational webinars and podcasts, contributions to consensus documents/guidelines documents, resources to learn from best practices and a scientific research programme among others consisting of a large patient registry. Since its inception, the PERT Consortium has more than 100 institutions registered, and it still growing, also across the US borders. Notably, for several reasons, some aspects of PERT recommendations and activities may not be fully transferable to the European setting, notably organization of care and treatment algorithms (*Figure 3*). *Firstly,* European and US healthcare systems differ considerably in their structure, funding, and outcomes. European nations generally favour universal healthcare, with publicly funded systems that aim for broad access and cost control, while the US operates a more market-based system with a mix of public and private insurance. Even within Europe, healthcare systems exhibit diversity in their organization, financing, and delivery of care, despite the common goals mentioned before. These differences stem from historical, cultural, and political factors and are notable for changing over time. In general, universal equitable access to healthcare is prioritized, preferably delivered by public healthcare providers. A common focus on strategies to control healthcare costs leads to high usage of generics, strict health technology assessment procedures, and stringent managed entry agreements for new drugs. Compared to the US setting, the European medical device regulation (MDR) has more extensive regulatory requirements for the equivalence pathway. The General Data Protection Regulation (GDPR) strictly regulates how hospitals and other healthcare providers handle personal data, including health information. Communication between hospitals, including sharing patient data, is subject to these regulations, requiring secure and appropriate methods for data exchange. This 'European perspective' necessitates a more generalist approach to healthcare initiatives such as EXPERT-PE than would have been the case in the US, to allow widespread applicability and acceptability, with a clear focus on health equity and overcoming differences between European countries and settings. Notably, while the ESC/ERS guidelines is commonly regarded as leading for the European region,[12] numerous other guidelines, practice documents and consensus statements have been published and are applied, among others from almost all individual countries.

Secondly, the healthcare provider primarily responsible for the immediate and long-term care of patients with acute PE differs between European countries: cardiology is the speciality most often involved in Germany, Denmark and Poland, while it is pulmonology and internal medicine in the Netherlands and Spain. In Italy, PE care is mainly provided by cardiologists and internists while in France, key roles are played

Table 1 Variables to be included in the minimal and extended dataset for pulmonary embolism

Minimal dataset	Extended dataset
Timepoint 1: Initial Assessment and Acute Treatment	
Demographics (date of birth, age, sex, ethnicity, height, weight, body mass index, smoking status)	ICU stay
Risk Stratification (low/intermediate-low/intermediate-high/high)	Hospital admission source (direct vs. transfer from outside facility); treatment before transfer (if transferred from other facility)
EXPERT-PE (team notification/team meeting: timing/figures involved/decision)	Enrolled in PE clinical trial
Comorbidities and risk factors	
Cancer (active, cured or >6 mo remission)	Known inherited thrombophilia
Previous personal DVT/PE; family history VTE	COVID-19
Recent hospitalization, major surgery, major trauma (<4 weeks from hospital admission), indwelling lines present	Limited mobility
Pregnancy or postpartum	Previous stroke
Active major bleeding	Known APLS
Chronic cardiac disease (previous AMI/heart failure), kidney disease (dialysis/other stage), pulmonary disease (COPD/already on oxygen support at home), diabetes (on/off insulin)	Oral contraceptives/hormone replacement therapy
Known pulmonary hypertension (CTEPH/other type)	Current arterial antihypertension treatment
IVC filter (already present/inserted <48 h from PE diagnosis)	Other major risk factors (inflammatory bowel syndrome/long distance travel/leg injury/HIV/varicose veins/sepsis/major burn/HIT)
Clinical symptoms, vital signs	
Temperature/heart rate/systolic blood pressure	Diastolic blood pressure
Respiratory rate, O2 saturation (room air/oxygen support/PaO2/FiO2)	Dyspnoea (modified Borg score), chest pain, dizziness/ light-headedness/syncope (with duration in days)
Cardiac arrest/CPR administered	Level of consciousness (GCS)
Non-invasive positive pressure ventilation/mechanical ventilation	
Contraindication to anticoagulation/thrombolysis (specify type)	
Medication history	
Anticoagulant (type/dose); antiplatelet agents (type/dose)	Hormonal therapy; Active antineoplastic therapy
Pretreatment laboratory and imaging parameters	
Laboratory results (D-dimer, creatinine, lactate, haemoglobin, platelets, troponin, BNP, aPTT)	
CT imaging (right heart strain, RV/LV ratio, thrombus location, clot in transit)	
Echocardiogram and/or POCUS [RV function and size (qualitative), clot in transit]	
Timepoint 2: Acute management and interventional parameters	
Anticoagulation type, dose, times (initial/subsequent)	Escalation of care (advanced therapies)
Vasopressors (type, dose, time, duration)	Surgical embolectomy (procedure date, duration)
Systemic thrombolysis (date, time, agent)	ECMO/mechanical circulatory support (type/timing/duration)
Catheter directed local thrombolysis or catheter directed thrombectomy (device, drug, dose, duration)	Angiographic improvement; Thrombus removal; Device/technical success; Blood loss estimated
Intraprocedural haemodynamic and procedural outcomes: RA/PA pressures, PA O2 saturation	Acute outcome (clinical/procedural success)
SAE within 48 h (haemoglobin, major bleeding, transfusion, stroke, arrhythmia, vasopressor requirement, haemoptysis, cardiac arrest, access site complication, renal replacement therapy, clinical deterioration, recurrent PE, death)	
Timepoint 3: Postprocedural In-hospital Monitoring and Management	
Adjunct therapy [IVC filter (from 48 h until hospital discharge), other]	Posttreatment laboratory results (creatinine, haemoglobin, platelets, aPTT)

Continued

Table 1 ***Continued***

Minimal dataset	Extended dataset
Vital signs (NEWS calculation: HR, BP, O2 saturation, respiratory rate, consciousness, temperature)	
Posttreatment echocardiogram [RV function and size (qualitative), RV/LV ratio]	
Timepoint 4: Discharge Parameters	
Hospital discharge or all cause death/fatal PE/fatal bleeding (date)	General (disposition location/facility, follow-up plan; educational materials)
Functional status on day of discharge (dyspnoea score; NYHA class; supplemental oxygen)	
Timepoint 5: Postdischarge Management	
Anticoagulation at follow-up (anticoagulant drug, dose, duration, adherence, change from previous)	Pharmacological education provided
VTE follow-up and intercurrent events, specify dates (functional status; recurrent VTE events; bleeding events; IVC filter management; thrombophilia evaluation; ongoing risk factor assessment)	Follow-up visits conducted [time frame (1, 3, 6, 12 mo); provider type; virtual/in-person]
Follow-up laboratory results/imaging, specify dates (repeat CTA/echocardiogram; venous duplex)	Symptom screening; Exercise function (CPET); Arterial perfusion (planar or SPECT V/Q)
Symptoms at follow-up [dyspnoea (Borg scale); chest pain; lower extremity edema]	Further intervention
CTEPD/CTEPH evaluation at ≥3 mo (per SEARCH algorithm)	
Readmission: VTE-related/specific indication	

by cardiologists and pulmonologists but also vascular medicine and internists. In other countries such as UK and Ireland, haematologists may take the lead. Implicit differences between those medical specialities and their priorities and views on optimal care as well as their current European guidelines are to be considered when introducing EXPERT-PE care teams across Europe and providing best practices for their successful implementation and uptake. Given the diversity of contributors across countries, a 'European perspective' demands a flexible and generalist approach to bring together complementary skills and motivations: one that ensures broad applicability across different national contexts, emphasizes health equity, and allows each country or region to adapt the EXPERT-PE framework to its existing structures.

Research priorities

Playing a central role in the management of severe-PE and being the keeper of the highest quality of care for these patients, EXPERT-PE care teams have the social and academic obligation to take the lead in clinical research in the area. This research should primarily focus on improving (prehospital) triage, better patient selection for reperfusion treatment, enhanced efficiency and quality of EXPERT-PE care team deployment, and optimal organisation of care in the European setting, considering regional and national differences. Scientific endeavours in this area should primarily be in the setting of well-designed randomized controlled trials, but national registries to capture the applied care and its outcomes in daily practice are also relevant. Indeed, the evolving landscape of healthcare in patients with acute PE requires monitoring the impact of modern PE treatment and underscores the need for standardization of data collection in routine care, which is essential to understand its impact. A list of critical evidence gaps with a matching comprehensive set of standardized data elements and data collection points was recently proposed by PERT.[11] Standardized collection of data provides a foundation for productive research and contributes to answering urgent unmet needs in PE management. Inspired by the core dataset proposed by PERT, and mindful of the feasibility of collecting large amounts of data in daily practice, the authors of this review voted on the relevance of the individual data elements; variables that achieved at least 75% agreement amongst authors were retained in the minimal dataset. Therefore, 5 of the 42 proposed variables were moved to the extended dataset. Details of the final standardized dataset (structure and variables) are reported in *Table 1*.

Conclusion

In parallel with contemporary advances in reperfusion treatment of acute severe PE, the establishment of multidisciplinary PE care teams with a clear organization, standardized and streamlined activation process, and personalized decision-making based on evidence-based treatment protocols may optimize the management of PE patients presenting with overt or imminent cardiopulmonary decompensation. At a given hospital and for a given patient with potentially life-threatening PE, the therapeutic strategy decided upon through consensus within the PE-dedicated team increases the chances of success by making the best use of local expertise and resources, and of regional transfer networks. The EXPERT-PE study group, established by the ACVC in collaboration with the ESC Working Group on Pulmonary Circulation and Right Ventricular Function, serves as a central reference for EXPERT-PE care teams across Europe and beyond, fostering collaboration, harmonizing protocols and scientific research, and overall advancing care for patients with acute PE.

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Data availability

The anonymized data that support the findings of this work are available on request from the corresponding author.

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